

Homework 5

Due: Friday, March 18

All homeworks are due at 11:59 PM on Gradescope.

Please do not include any identifying information about yourself in the handin, including your Banner ID.

Be sure to fully explain your reasoning and show all work for full credit.

LaTeX tips

Here are some LaTeX commands that might help you

$$\begin{array}{llll} \backslash equiv & \equiv & a \backslash mid b & a \mid b \\ & & \backslash gcd(a, b) & gcd(a, b) \\ & & a \backslash equiv b \backslash pmod{m} & a \equiv b \pmod{m} \end{array}$$

LaTeX *is* required. You should be using the `cs22` document class. Look for a LaTeX guide [here](#) for more information.

Problem 1

Consider a sequences of numbers $a_1, a_2, a_3, \dots$ where $a_n = 50 + n^2$ for $n = 1, 2, 3, \dots$. The first few values are

$$a_1 = 51, \quad a_2 = 54, \quad a_3 = 59, \quad a_4 = 66, \quad a_5 = 75, \quad \dots$$

- a. *Warm-up.* Compute a_{15}, a_{16}, a_{167} , and a_{168} . Check that you got the values:

$$a_{15} = 275, \quad a_{16} = 306, \quad a_{167} = 27939, \quad a_{168} = 28274$$

- b. Use the Euclidean algorithm to calculate the following values, writing out every intermediate step. *Providing the answer without any steps will yield no credit.*

i. $\gcd(a_{15}, a_{16}) = \gcd(275, 306)$.

ii. $\gcd(a_{167}, a_{168}) = \gcd(27939, 28274)$.

- c. Using part (b.i) from above, and using the extended Euclidean algorithm, solve the following congruence, writing out every intermediate step. *Providing the answer without any steps will yield no credit.*

$$x \text{ such that } 275 \cdot x \equiv 1 \pmod{306}.$$

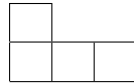
- d. We are interested in the value of $\gcd(a_n, a_{n+1})$ for different values of n . From part (b), we know that this can be different values.

Let's compute two more examples. Use a calculator¹ to compute the following:

- i. $\gcd(a_{102}, a_{103}) = \gcd(50 + 102^2, 50 + 103^2)$.
 - ii. $\gcd(a_{368}, a_{369}) = \gcd(50 + 368^2, 50 + 369^2)$.
- e. Verify that your answers from parts (b) and (d) satisfy
- $$\gcd(a_n, a_{n+1}) = \gcd(n^2 - 100n, 2n + 1) = \gcd(n - 100, 201)$$
- f. Using the Euclidean algorithm and algebraic manipulation, show that the property in part (e) holds true for all n .
- g. For which n is $\gcd(a_n, a_{n+1})$ the greatest, and what is the maximum value? Find and classify *all* n .

Problem 2

- a. Prove that if an integer n is odd, then either $n + 1$ or $n - 1$ is divisible by 4.
- b. Prove using strong induction that for any odd integer $n \geq 3$, an $n \times n$ chessboard with the center square removed can be tiled by these 4-square L-tiles:



Hint: Show this for $n = 3$ and $n = 5$ first as your base cases. Then, for every other odd number n , can we add a 2-thick border to the tiling for $n - 4$ using our L-tiles?

Problem 3

Show that if m is a positive integer greater than 1 and

$$ac \equiv bc \pmod{m},$$

then

$$a \equiv b \pmod{\frac{m}{\gcd(c, m)}}.$$

Hint: Use the definition $a \equiv b \pmod{m} \Leftrightarrow m \mid a - b \Leftrightarrow \exists k \in \mathbb{Z} \text{ s.t. } mk = a - b$.

¹You are welcome to use a site such as [wolframalpha.com](https://www.wolframalpha.com) and simply enter a mathematical expression, such as ' $\gcd(50 + 102^2, 50 + 103^2)$ ', into the search box. You are welcome to use this to *verify* your solutions to part (b), but you should show your steps.

☕ Problem 4 (Mind Bender — *Extra Credit*)

Mind Benders are extra credit problems intended to be more challenging than usual homework problems and are an exploration into a topic not covered in lecture. This week, we have an exploration into *pseudoprimes* and *Carmichael numbers*.

From Fermat's Little Theorem, we know that for prime p , $a^p \equiv a \pmod{p}$ is true for $p \nmid a$. One might wonder if the converse of this is true. If $a^n \equiv a \pmod{n}$, does n have to be a prime? This is unfortunately not true.

Example

Take $n = 341$ and $a = 2$. $2^{10} = 1024 \equiv 1 \pmod{341}$ and $2^{341} = (2^{10})^{34} \cdot 2 \equiv 2 \pmod{341}$. But $341 = 11 \cdot 31$, which is not a prime.

When this happens, we call n a *pseudoprime* to the base a if n is composite but $a^n \equiv a \pmod{n}$. So, 341 is a pseudoprime to the base 2.

Are there composite numbers n that are pseudoprime to *all* bases a ? That is, for all $a \in \mathbb{Z}$, $a^n \equiv a \pmod{n}$ is true.

Example

Yes! 561 is one such number.

These numbers are called *Carmichael numbers*².

- a. Prove that any n satisfying the following conditions is a Carmichael number³:
 - i. n is odd.
 - ii. n is squarefree. That is, each prime in n 's prime factorization appears only once; there are no powers of primes that divide n .
 - iii. For every prime p dividing n , we have that $(p - 1) \mid (n - 1)$.

Hint: By the Chinese Remainder Theorem, if $a \equiv b \pmod{p_i}$ for distinct primes p_i , then $a \equiv b \pmod{N}$, where N is the product of the p_i 's.

- b. Prove that a Carmichael number cannot be the product of two distinct primes.

²Named after [Robert Daniel Carmichael](#).

³This actually goes both ways. If n is a Carmichael number, n satisfies those conditions. We only prove the backward direction. This is called *Korselt's Criterion*.